RE07 Global Infrastructure Investment and Management

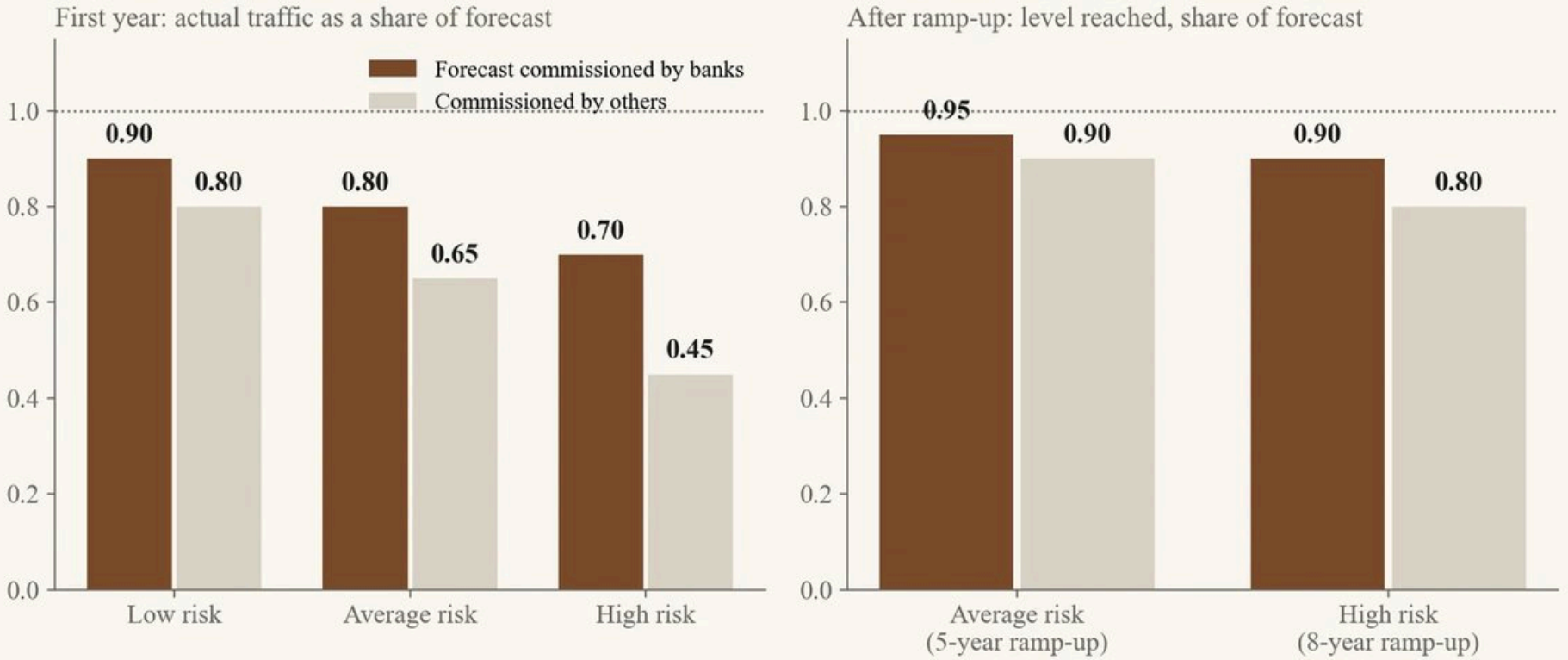
6. Performance and Valuation

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Lecture Outline

Forecasts and the base case
Measuring the performance of unlisted funds
Discount rates and the valuation of regulated assets
Data centres: betas, lives and residual values
Three models of measuring an asset that does not trade
Case Study: Index Construction and Loan Performance (Andreas Jobst, Central Bank of the UAE)

Traffic Forecasts and Ramp-Up



Source: Lemp and Kockelman (2009), pre-print of Transportation Research Record 2132, pages 2 to 4, summarising Bain and Wilkins (2002) and later Standard & Poor's studies of 32, 68 and 87 toll roads.

The record: first-year traffic on new toll roads averaged 0.73 to 0.77 of forecast in four Standard & Poor's studies

Where it bites: about three quarters of first-year forecasts were overestimated by 10 per cent or more

Ramp-up: two years for low-risk projects, five for average and eight for high-risk, with the last two settling below forecast

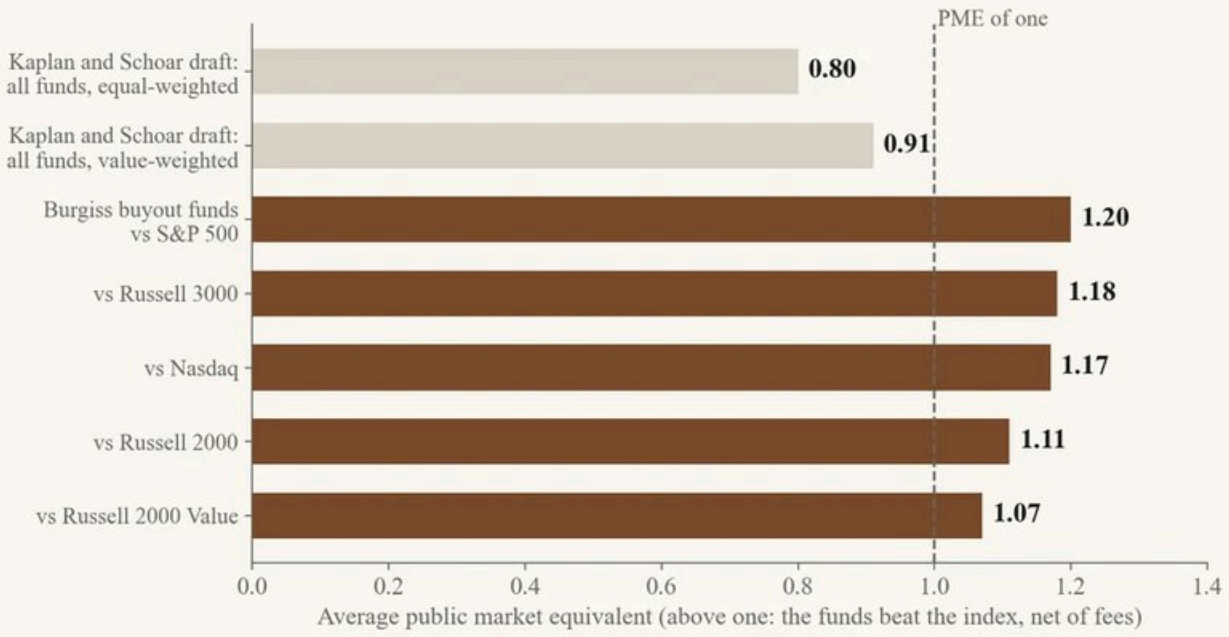
Who commissioned it: forecasts ordered by banks erred less than those ordered by others, 0.80 against 0.65 for average-risk projects

Measures of Fund Performance

Measure	What it states	What it leaves out
Internal rate of return	The annual rate that sets the discounted net cash flows to zero	What the money could have earned elsewhere; the manager's estimate of any remaining value
Multiple of invested capital	Distributions plus remaining value for each unit contributed	Timing: a multiple of 1.8 over five years and over fifteen is the same number
Public market equivalent	Value of distributions relative to value of contributions, both compounded at an index (Kaplan and Schoar)	The choice of index and its beta; a ratio and not a rate
Direct alpha	The internal rate of return of the index-compounded flows, an annual excess return (Gredil, Griffiths and Stucke)	The same dependence on the benchmark; the paper itself was not read

Kaplan and Schoar (2005), NBER conference draft of 6 April 2003, pages 9 and 10; Harris, Jenkinson and Kaplan (2012), NBER Working Paper 17874, page 10; Gredil, Griffiths and Stucke (2014 working paper) as described in secondary sources.

Private Equity Performance across Data Sets and Benchmarks



Early data: using Venture Economics data, the average private equity fund earned 0.80 of what the S&P 500 would have (0.91 weighting by commitment)

Later data: Burgiss cash flows give U.S. buyout funds an average of 1.20, an outperformance of 20 to 27 per cent over a fund's life

The benchmark: the buyout figure falls from 1.20 against the S&P 500 to 1.07 against small-cap value, still above one

Venture capital: above one for most 1990s vintages and below one for the 1999 to 2002 vintages

Kaplan and Schoar (2005), NBER conference draft of 6 April 2003, page 10 (Venture Economics, 776 funds); Harris, Jenkinson and Kaplan (2012), NBER Working Paper 17874, pages 16 to 19 (Burgiss, U.S. buyout funds). Different samples and periods.

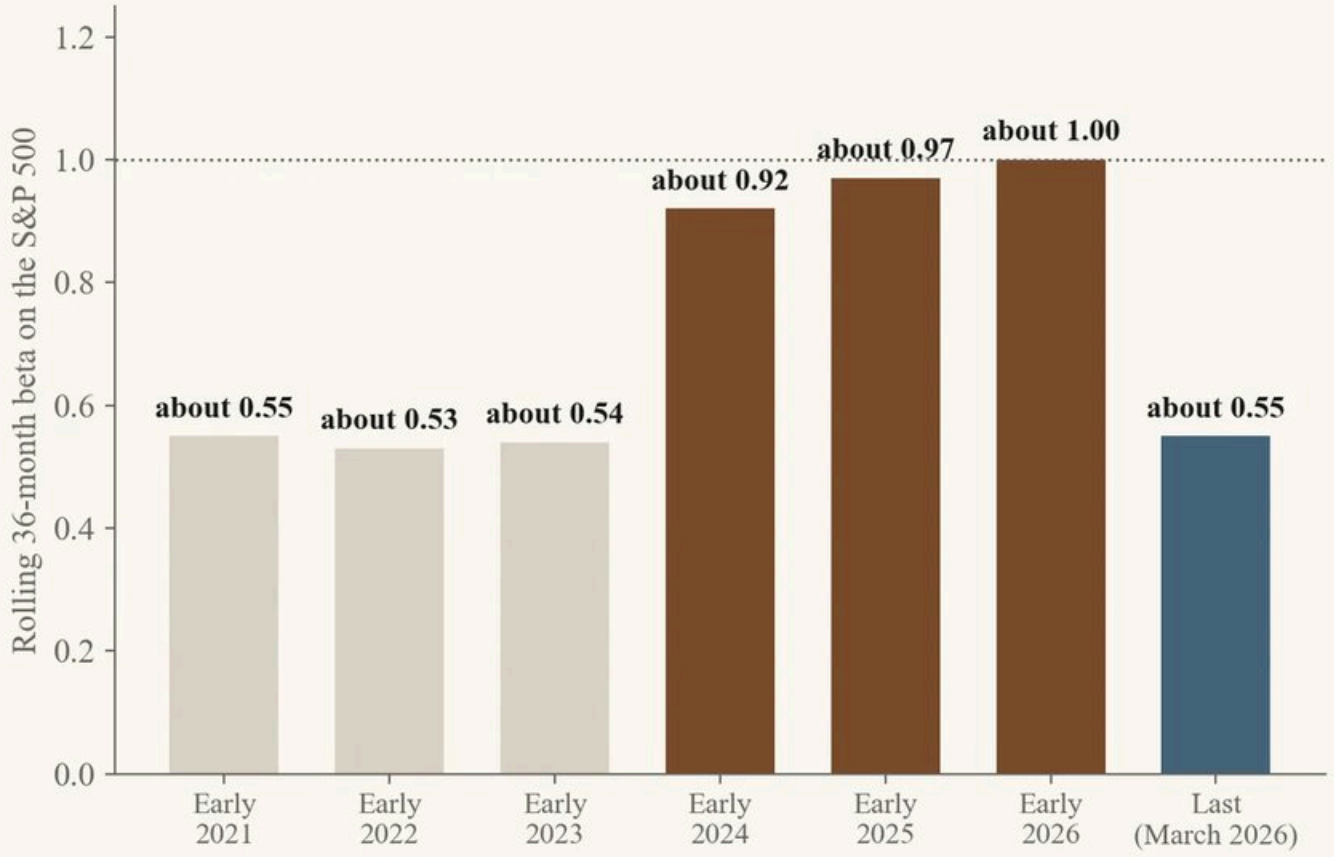
Allowed Returns in Regulated Water

Parameter (CPIH-real)	Ofwat final determination	CMA provisional findings	CMA final determination
Market data cut-off	30 September 2024	30 June 2025	30 November 2025
Gearing	55%	55%	55%
Risk-free rate	1.52%	2.49%	2.40%
Equity beta	0.59 to 0.65	0.61 to 0.76 (0.68)	0.61 to 0.70 (0.66)
Total market return	6.68 to 6.98%	6.70 to 7.30% (7.00%)	6.70 to 7.20% (6.95%)
Return on equity (point)	5.10%	5.90%	5.70%
Return on debt	3.15%	2.98%	2.97%
Weighted cost of capital (appointee)	4.03%	4.29%	4.20%

The CMA's decision raised the return on equity by 60 basis points over Ofwat's, and the weighted cost by 17 basis points on the appointee basis and 23 on the wholesale basis.

Oxera (27 March 2026), CMA PR24 water redeterminations, Table 2.1, page 2, on the CMA's final determinations of 10 March 2026. Point estimates in brackets. A consultancy note; the CMA's report was not read.

Data-Centre Betas



The reading: the beta of listed data-centre property companies on the S&P 500 was about 0.5 from 2021 to 2023 and about 1 through 2024 and 2025

The last point: it fell back to about 0.55 in March 2026, so the rise is not yet a settled level

The paper's reading: data centres look less like defensive infrastructure and more like part of the same systematic risk as AI stocks

For valuation: a comparator beta of 0.5 and one of 1.0 give a weighted cost of 6.35 and 8.10 per cent, and values 16 per cent apart (slide 15)

Van Nieuwerburgh (2026), NBER Working Paper 35865, Figure 3 on page 14 and text on page 13. Values read by eye from the figure, accurate to about 0.05.

Asset Lives and Reported Earnings in Data Centres

Company	Change from 1 January 2025	Stated effect, first half of 2025
Meta	Most servers and network assets: useful life lengthened to 5.5 years	Depreciation lower by $1.58 billion; net income higher by $1.34 billion
Amazon	A subset of servers and networking equipment: useful life shortened from six years to five	Depreciation and amortisation higher by $497 million; net income lower by $379 million

Buildings and power systems outlast chips: capacity commissioned in 2025 and 2026 may need substantial GPU replacement by about 2030 while the buildings remain usable (Van Nieuwerburgh, 2026, page 24).

Meta Platforms and Amazon.com, Forms 10-Q for the quarter ended 30 June 2025, Note 1 (Use of estimates); Van Nieuwerburgh (2026), NBER Working Paper 35865, pages 12, 15 and 24.

Measuring an Asset That Does Not Trade

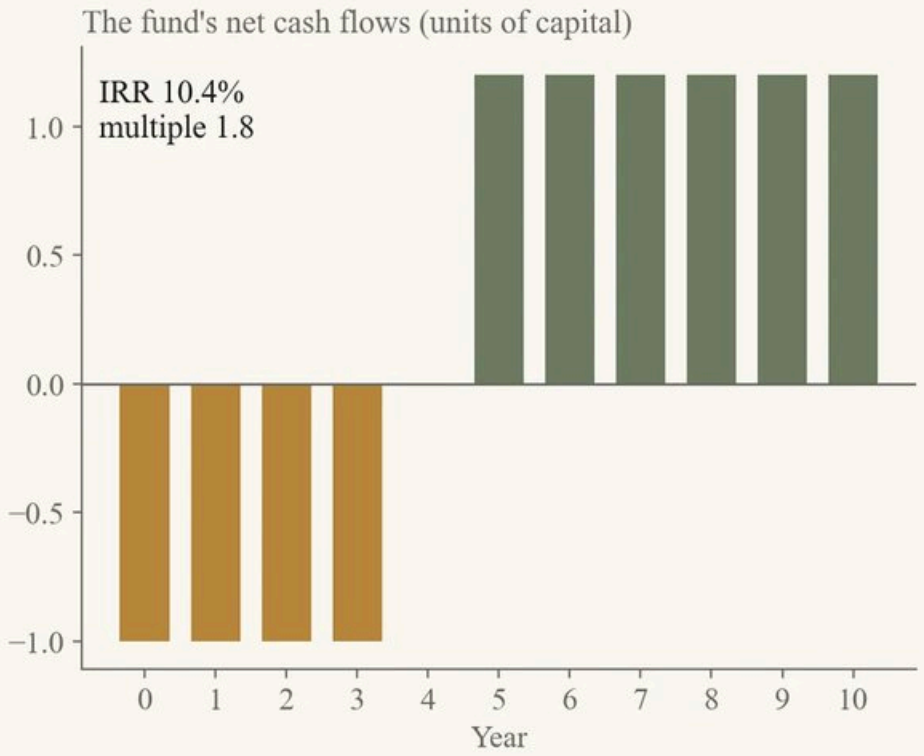
Performance against the market: the public market equivalent and the direct alpha: Kaplan and Schoar (2005); Gredil, Griffiths and Stucke

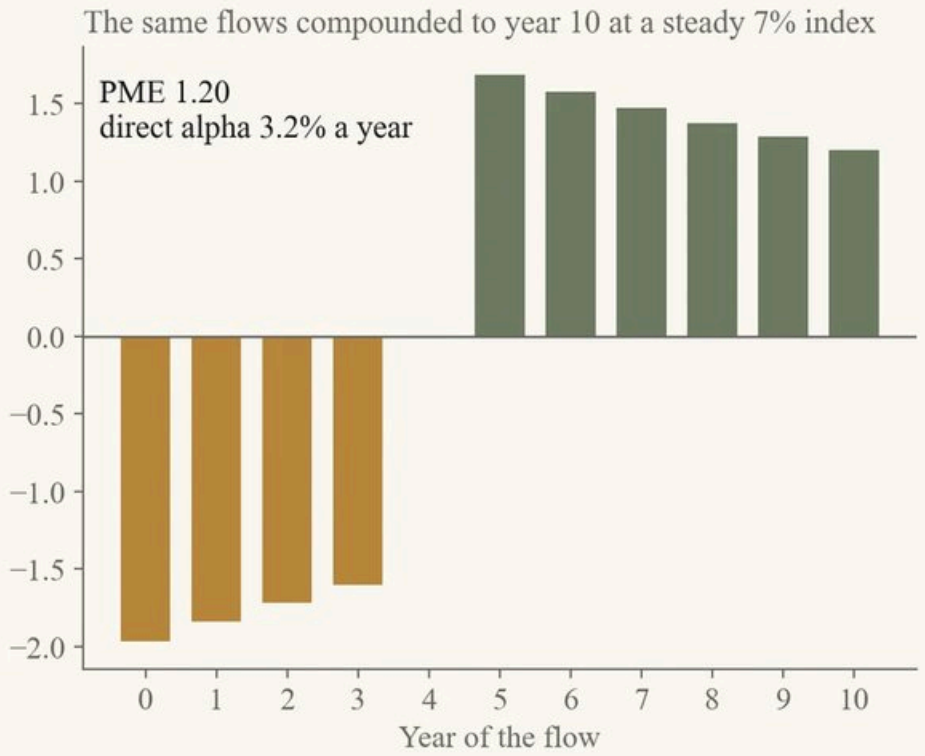
Reported values: the smoothing of appraisal-based returns: Getmansky, Lo and Makarov (2004)

Discount rates: the cost of capital of an asset with no traded price: listed comparators, the CMA's determination for water, and the cost of illiquidity

The common problem: the asset does not trade, so a return or a value is inferred from cash flows, appraisals or comparators, and each inference has a known bias

Kaplan and Schoar (2005): The Public Market Equivalent





Source: Kaplan and Schoar (2005), NBER conference draft of 6 April 2003, pages 7 and 9; Gredil, Griffiths and Stucke as described in secondary sources. A generic fund, the settings of pme.html; not calibrated to any fund.

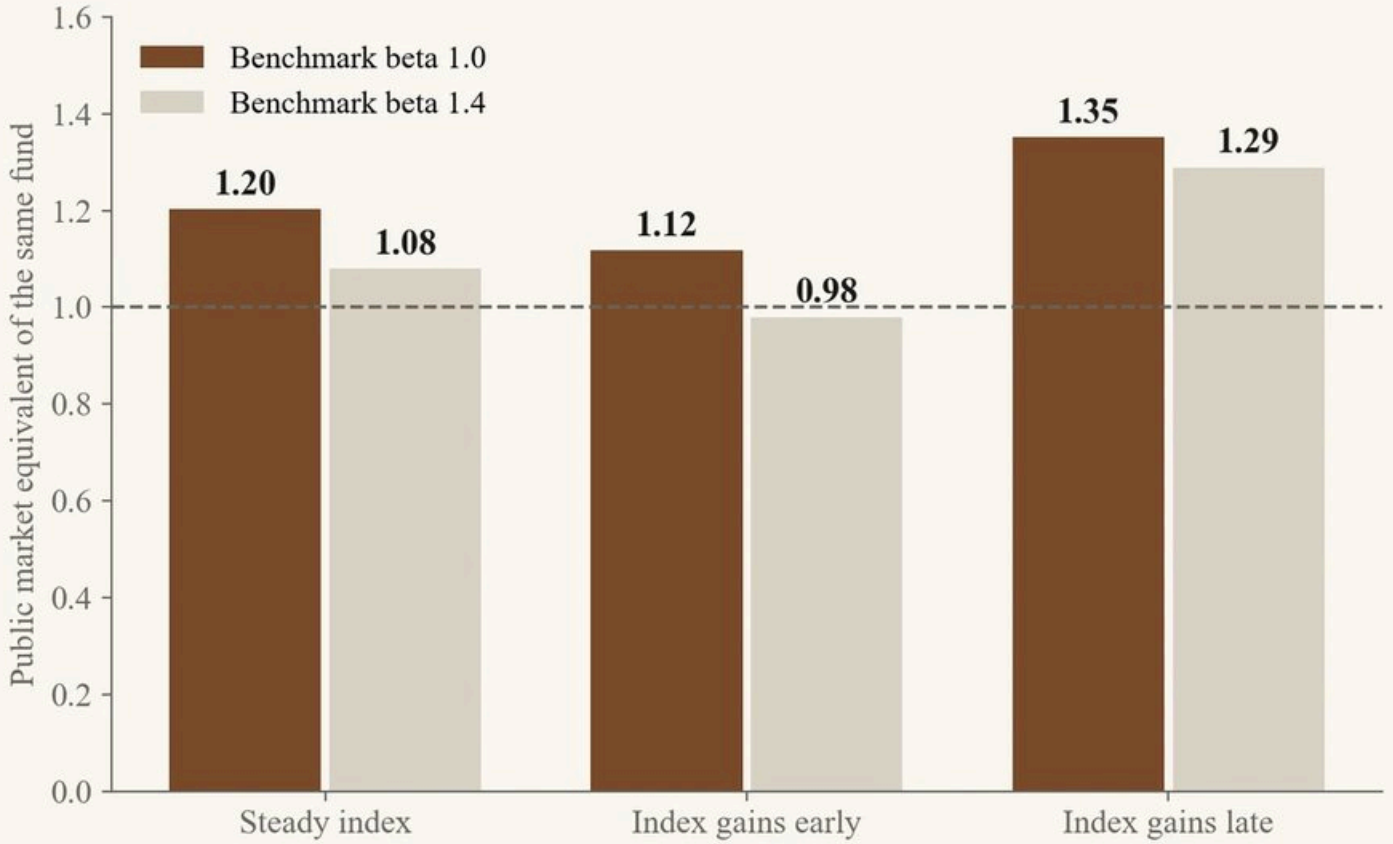
The idea: compound each contribution and each distribution at an index, and compare the value of what came back with the value of what went in

The result: a fund with a multiple of 1.8 called over four years and returned over years 5 to 10 has an IRR of 10.4 per cent and, against a steady 7 per cent index, a PME of 1.20

Direct alpha: the same compounded flows give an annual excess return of 3.2 per cent, with the same sign as the PME

The condition: the fund must be fully realised, or its last reported value stands in for a sale

The Verdict Depends on the Benchmark



Timing: with the index gaining early, the same fund's PME falls to 1.12; gaining late, it rises to 1.35

Beta: scaling the benchmark to a beta of 1.4 cuts the PME to 1.08 against a steady index

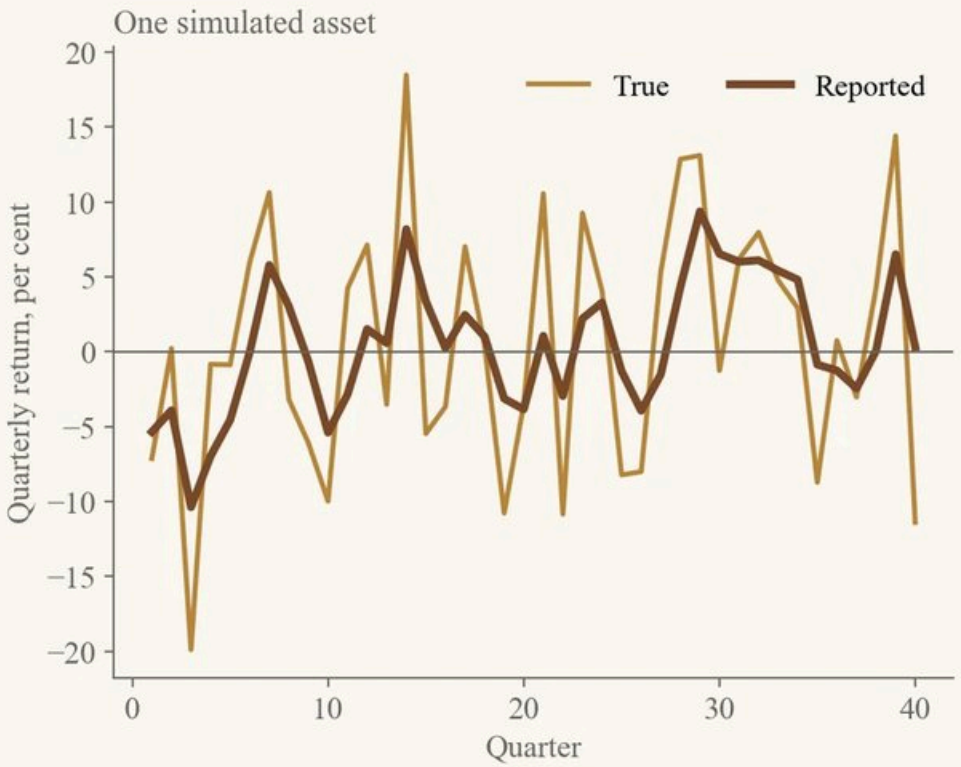
Data: the method gives 0.80 on one data set and 1.20 on another (slide 5)

The statement to check: any claim that a fund outperformed needs the index, its beta, the dates and the share of value that is unrealised

Kaplan and Schoar (2005), draft of 6 April 2003, page 9; Harris, Jenkinson and Kaplan (2012), pages 16 to 19. Generic settings of pme.html; not calibrated to any fund.

Getmansky, Lo and Makarov (2004): Smoothed Returns





Source: Getmansky, Lo and Makarov (2004), NBER Working Paper 9571, pages 18 to 22 (equations 21 to 33). Generic settings of smoothing.html; one random path; not calibrated to any asset.

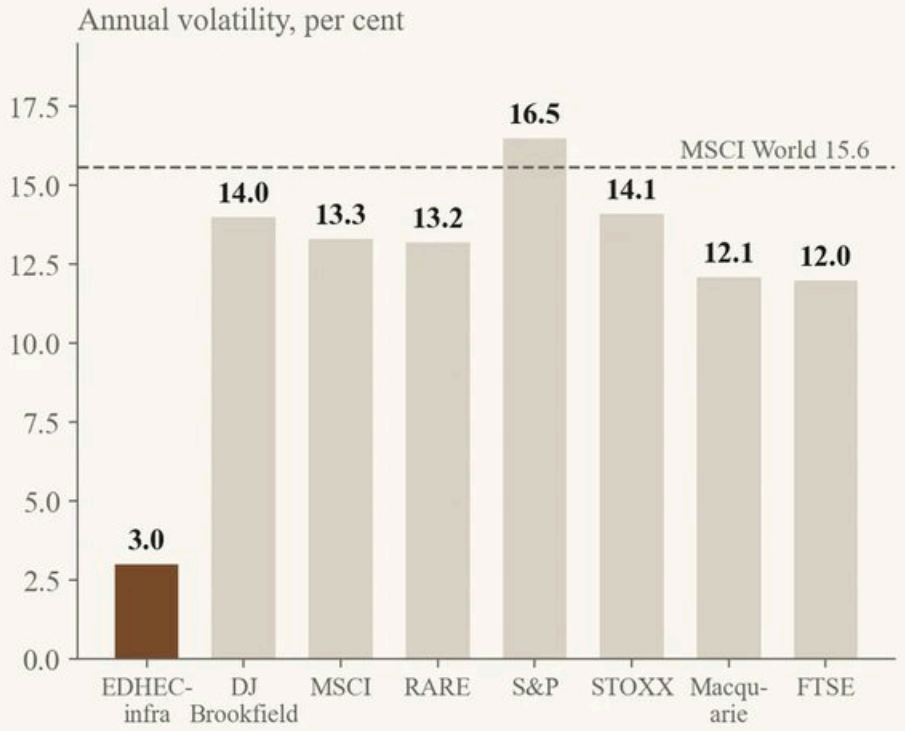
The model: the reported return is a weighted average of the true returns of the current and the last k quarters, with weights summing to one

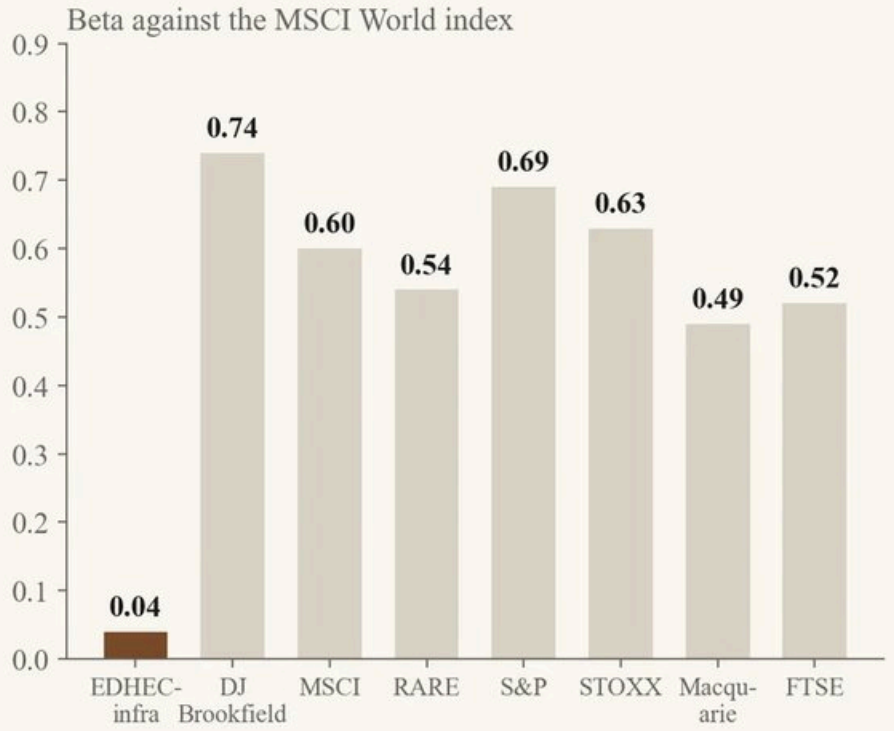
The example: weights of 0.4, 0.3, 0.2 and 0.1 on the current and three earlier quarters

The effect: reported volatility is 55 per cent of the true figure, the Sharpe ratio 1.83 times as high, and the same-quarter beta 0.32 against a true 0.80

What survives: the average return and, over five years, nearly all of the volatility (97 per cent of the true figure)

Smoothing in Unlisted Infrastructure Indices





Source: Lambrev (2019), Naše gospodarstvo / Our Economy 65(3), Table 8, page 35: quarterly data, 2007 to 2018 according to the text, against the MSCI World index. Smoothing model: Getmansky, Lo and Makarov (2004).

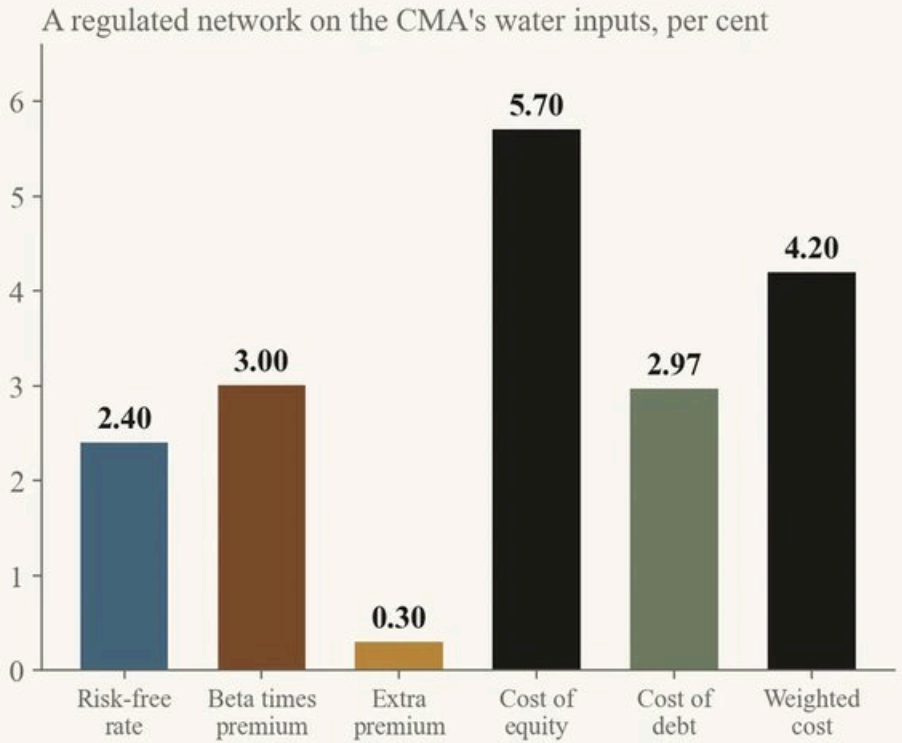
EDHECinfra, unlisted: volatility 3.0 per cent, beta 0.04 and a Sharpe ratio of 5.33

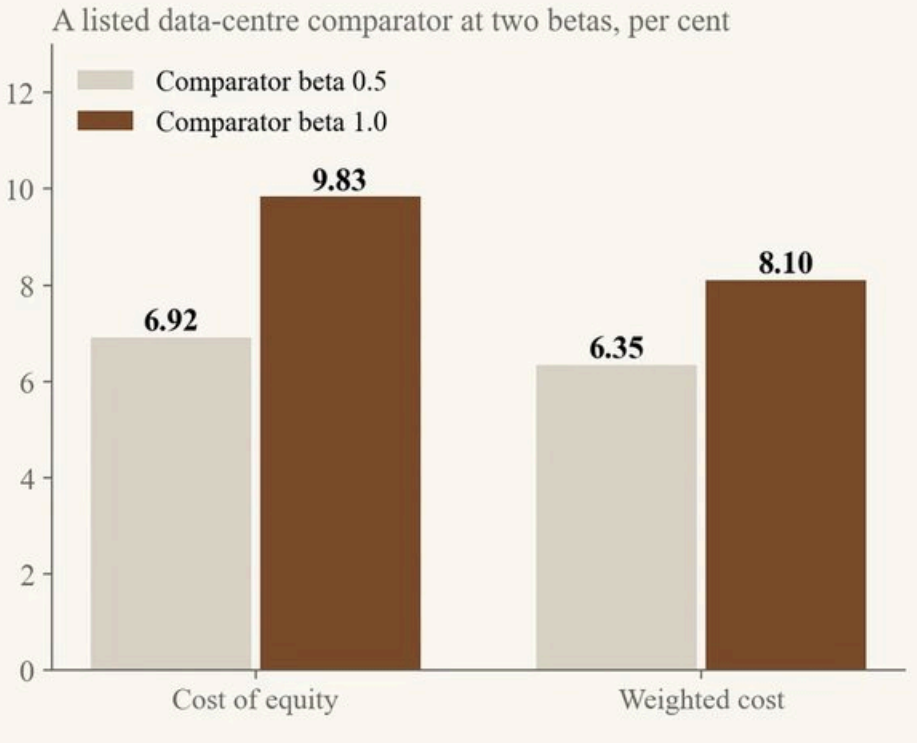
Listed indices: volatility 12.0 to 16.5 per cent, betas 0.49 to 0.74 and Sharpe ratios 0.10 to 0.69

Two readings: the paper reads the gap as listed indices misrepresenting infrastructure; the smoothing model offers a second reading, which the paper does not test

The correction: desmoothing raises measured volatility and beta, and needs an estimate of the lag

Building a Discount Rate from Listed Comparators





Source: Oxera (27 March 2026), Table 2.1, page 2, on the CMA's determination; generic data-centre inputs; the extra 0.30 points is mine. Ang, Papanikolaou and Westerfield (2013), NBER Working Paper 19436.

Unlever and relever: the equity beta of a listed comparator is stripped of its gearing and put back at the asset's own

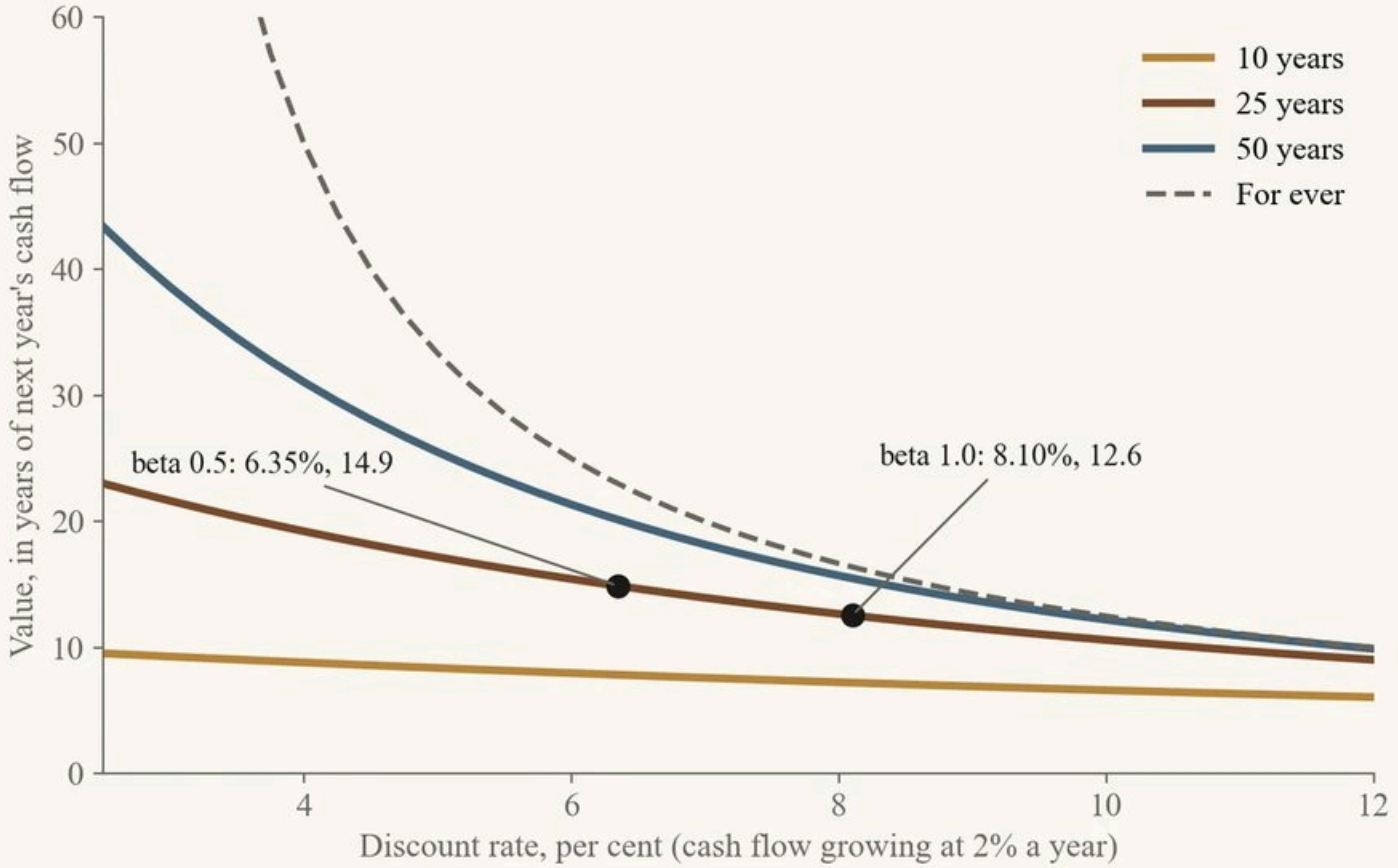
A regulated build-up: risk-free rate 2.40, beta 0.66 on a premium of 4.55 and 0.30 for the CMA's choice give a cost of equity of 5.70 per cent; weighted with debt at 2.97, 4.20

Listed data centres: a comparator beta of 0.5 gives a weighted cost of 6.35 per cent and 1.0 gives 8.10, on the same other inputs

Illiquidity: an extra premium can be added; its size depends most on the uncertainty about how long the asset cannot be sold (Ang, Papanikolaou and Westerfield, 2013)

A dated input: the median ten-year government yield in rich countries passed 4 per cent in 2026, about five times its 2015 to 2021 average (The Economist, 16 September 2026)

Value as a Function of the Discount Rate



Sensitivity: a 25-year cash flow growing at 2 per cent is worth 14.9 years of next year's cash at 6.35 per cent and 12.6 at 8.10, 16 per cent less

Length: the loss is 8 per cent for a 10-year flow, 16 for 25 years and 23 for 50, and 29 for a flow that never ends

One point: for a perpetuity, a rate one point lower raises value by 30 per cent

Economic life: a cash flow that may end in ten years is worth 7.8 at 6.35 per cent, against 14.9 for 25 years

Growing annuity of models_w6.py at generic inputs (growth 2 per cent); betas of 0.5 and 1.0 from Van Nieuwerburgh (2026), Figure 3, read by eye. Not a valuation of any asset.

Three Models Compared

	Kaplan and Schoar (2005); direct alpha	Getmansky, Lo and Makarov (2004)	Discount rate model (CMA 2026; textbook)
What it corrects	The IRR, for the opportunity cost of the money	Reported risk and return, for the lag in valuations	The price of an asset with no price
Main input	A benchmark, its beta and the dates of the flows	Weights on past true returns	A comparator beta, gearing and a premium
What can go wrong	A benchmark of the wrong risk; a final value that is an estimate	Weights that move; appraisers who respond to news	A comparator beta that moves; a premium that is a guess
What the slides show	PME 0.80 or 1.20 by data set; 1.12 to 1.35 by index path	Volatility 55 per cent of true; beta 0.32 for a true 0.80	Weighted cost 6.35 against 8.10; values 16 per cent apart
A case could test it by	Fund cash flows and the benchmark	Quarterly values and the later sales	The rate in a bid and its comparators

Kaplan and Schoar (2005); Gredil, Griffiths and Stucke; Getmansky, Lo and Makarov (2004); Oxera (2026) on the CMA; Ang, Papanikolaou and Westerfield (2013). Interactive pages in web/.

Case Study: Index Construction and Loan Performance (Andreas Jobst, Central Bank of the UAE)

The question and the data: what is measured, for whom, and over what period

The index: which assets enter it, how they are valued and weighted, and what the index can and cannot say

Loan performance: default and recovery on unrated project loans, and how the data providers differ

A sector cut: what a water or renewables sample changes in the answer

Discussion: questions for the guest

Summary

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